



Mahindra University Hyderabad
École Centrale School of Engineering,
Minor-2 Examination Program: B.Tech
Branch: Computation & Mathematics Year: III Semester: I
Subject: Advanced Linear Algebra (MA3117)

Date: 31/10/2025

Time Duration: 1.5 Hours

Start Time: 02.00 PM

Max. Marks: 20

Instructions: All questions are compulsory.

Q 1:

5 marks

Using Givens rotations, find the matrix G satisfying $G \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \sqrt{3} \\ 0 \\ 1 \\ 0 \end{pmatrix}$.

Q 2:

5 marks

Let R_1 and R_2 be upper triangular matrices with positive diagonal entries. Find the unitary matrix V that satisfies the relation $R_1 = VR_2$.

Q 3 :

5 marks

Let $R = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 4 & 3 \\ 0 & 0 & 12 \end{bmatrix}$ be an upper triangular matrix in QR decomposition of matrix A . Find the 2-norm of each column of the matrix A .

Q 4:

5 marks

Let H be a Householder reflector with Householder vector $u \in \mathbb{R}^n$. Show that $\|Hx - y\|_2 = \|x - y\|_2$ for any vector y perpendicular to u and $x \in \mathbb{R}^n$.